Entropy and divergences. Consider a convex function

$$\begin{array}{rccl} \Phi: & \mathbb{R}_+^2 & \to & \mathbb{R}\cup\{+\infty\} \\ & (u,v) & \mapsto & \Phi(u,v) \quad \text{s.t. } \forall a\in\mathbb{R}_+ \ \Phi(au,av)=a\,\Phi(u,v) \text{ and } \Phi(1,1)=0. \end{array} \tag{1.12}$$

The Φ-entropy between two probability measures $\nu_1,\nu_2\in\mathcal{M}_1(S)$ is defined for any dominating measure γ (such that $\gamma\gg\nu_1$ and $\gamma\gg\nu_2$) by the formula

$$D_\Phi(\nu_1,\nu_2):=\int\ \Phi\left(\frac{d\nu_1}{d\gamma}(x),\frac{d\nu_2}{d\gamma}(x)\right)\ \gamma(dx). \tag{1.13}$$

By homogeneity arguments, the definition (1.13) does not depend on the choice of the dominating measure γ. Note that

$$\text{if}\quad \Phi(u,v)=\Phi_0(u,v):=\frac{1}{2}\,|u-v|\quad\text{then}\quad D_{\Phi_0}(\nu_1,\nu_2)=\|\nu_1-\nu_2\|_{\mathsf{tv}}. \tag{1.14}$$

Whenever $\Phi(1,0)=\infty$, the Φ-entropy coincides with the ϕ-divergence in the sense of Csiszár [30], i.e., we also have

$$D_\Phi(\nu_1,\nu_2)=\int\ \phi\left(\frac{d\nu_1}{d\nu_2}(x)\right)\ \nu_2(dx)\quad\text{where}\quad \phi(u):=\Phi(u,1)$$

and we have used the convention $D_\Phi(\nu_1,\nu_2)=\infty$ when $\nu_1\not\ll\nu_2$. Therefore, the Φ-entropy generalizes the classical relative entropy (a.k.a. Kullback-Leibler divergence or I-divergence). More precisely, choosing the functions

$$\Phi(u,v)=\Phi_1(u,v):=u\log(u/v)\quad\text{and}\quad \phi(u)=u\log u$$

we recover the classical formulae

$$D_{\Phi_1}(\nu_1,\nu_2)=\mathrm{Ent}(\nu_1\mid\nu_2):=\int\ \log\left(\frac{d\nu_1}{d\nu_2}(x)\right)\ \nu_1(dx)=\int\ \phi_1\left(\frac{d\nu_1}{d\nu_2}(x)\right)\ \nu_2(dx),$$

with the convex function with unique minimum $\phi_1(1)=0=\partial\phi_1(1)$ defined by

$$\phi_1(u)=u\log u-(u-1).$$

2. Statement of main results

This section introduces the Sinkhorn bridge construction, the associated time-inhomogeneous Sinkhorn semigroups, and states our main exponential stability results. Detailed proofs and refinements are deferred to later sections.

2.1. **Sinkhorn bridges.** The Schrödinger bridge distribution $\mathbb{P}$ defined in (1.4) can rarely be computed analytically. However, solutions can be approximated efficiently using the Sinkhorn algorithm, also referred to as the iterative proportional fitting procedure [32, 97, 98]. This procedure is defined sequentially for any $n\geqslant 0$ by a collection of probability distributions $(\mathcal{P}_n)$ and transition operators $(\mathcal{S}_n)$ given by

$$\mathcal{P}_{2n}=\lambda_U\times\mathcal{S}_{2n}\quad\text{and}\quad \mathcal{P}_{2n+1}=(\nu_V\times\mathcal{S}_{2n+1})^\flat \tag{2.1}$$

starting from $\mathcal{P}_0$ with $\mathcal{S}_0=\mathcal{Q}$, where, for a measure $\mu(dy)$ and a transition operator $\mathcal{S}(y,dx)$, the notation $(\mu\times\mathcal{S})^\flat$ denotes the 'reverse' measure

$$(\mu\times\mathcal{S})^\flat(d(x,y)):=\mu(dy)\mathcal{S}(y,dx).$$

The Markov transitions $\mathcal{S}_n$ in (2.1) are defined by the Sinkhorn iterations

$$\left\{\begin{array}{l} (\pi_{2n}\times\mathcal{S}_{2n+1})^\flat=\lambda_U\times\mathcal{S}_{2n}\quad\text{and}\quad \pi_{2n+1}\times\mathcal{S}_{2(n+1)}=(\nu_V\times\mathcal{S}_{2n+1})^\flat \\ \\ \text{with the distributions}\quad \pi_{2n}:=\lambda_U\mathcal{S}_{2n}\quad\text{and}\quad \pi_{2n+1}:=\nu_V\mathcal{S}_{2n+1}. \end{array}\right. \tag{2.2}$$

In the dual formulation, the bridge distributions $\mathcal{P}_n$ are often written as

$$\mathcal{P}_n(d(x,y)) = e^{-W(x,y)}\ \lambda_{U_n}(dx)\ \nu_{V_n}(dy), \tag{2.3}$$

with the initial potential functions $(U_0, V_0) = (U, 0)$ and potential functions (U_n, V_n) defined sequentially by the integral recursions

$$\begin{aligned} U_{2n+1} &= U_{2n} := U + \log \mathcal{Q}(e^{-V_{2n}}) \\ V_{2(n+1)} &= V_{2n+1} := V + \log \mathcal{R}(e^{-U_{2n}}). \end{aligned} \tag{2.4}$$

When $n \to \infty$, the sequence of distributions $\mathcal{P}_{2n}$ converges towards the Schrödinger bridge (1.4) from λ_U to ν_V (w.r.t. the reference measure $\mathcal{P}_0$) while the sequence $\mathcal{P}_{2n+1}$ converges towards the Schrödinger bridge from ν_V to λ_U (w.r.t. the reference measure $\mathcal{P}_1$).

Under mild integrability assumptions (see for instance Section 6 in [2]) the Schrödinger bridge distribution $\mathbb{P}$ solving (1.4) can also be computed from any Sinkhorn bridge reference distribution $\mathcal{P}_{2n}$, in the sense that for any $n \geqslant 0$ we have

$$\mathbb{P} = \underset{\mathcal{P} \in \Pi_{U,V}}{\arg\min}\, \mathrm{Ent}(\mathcal{P} \mid \mathcal{P}_{2n}). \tag{2.5}$$

The Sinkhorn transition operators can also be written in terms of the potential functions (U_n, V_n), namely,

$$\mathcal{S}_{2n}(x,dy) = \frac{\mathcal{Q}(x,dy)e^{-V_{2n}(y)}}{\mathcal{Q}(e^{-V_{2n}})(x)} \quad \text{and} \quad \mathcal{S}_{2n+1}(y,dx) = \frac{\mathcal{R}(y,dx)\ e^{-U_{2n+1}(x)}}{\mathcal{R}(e^{-U_{2n+1}})(y)}, \tag{2.6}$$

Moreover, Proposition 6.3 in [2] shows that the potentials can be expressed through the marginal flows as

$$V_{2n} = V_0 + \sum_{0 \leqslant p < n} \log \frac{d\pi_{2p}}{d\nu_V} \quad \text{and} \quad U_{2n} = U_0 + \sum_{0 \leqslant p < n} \log \frac{d\pi_{2p+1}}{d\lambda_U} \tag{2.7}$$

2.2. **Sinkhorn semigroups.** The Sinkhorn iterations naturally induce a family of time-inhomogeneous Markov semigroups. For each $n \geqslant 0$ define the forward-backward Gibbs-type transitions

$$\mathcal{S}^{\circ}_{2n+1} := \mathcal{S}_{2n}\mathcal{S}_{2n+1} \quad \text{and} \quad \mathcal{S}^{\circ}_{2(n+1)} := \mathcal{S}_{2n+1}\mathcal{S}_{2(n+1)}.$$

These kernels satisfy the fixed-point relations

$$\lambda_U \mathcal{S}^{\circ}_{2n+1} = \lambda_U \quad \text{and} \quad \nu_V \mathcal{S}^{\circ}_{2(n+1)} = \nu_V \tag{2.8}$$

and yield the marginal evolution equations

$$\pi_{2n} = \pi_{2(n-1)}\mathcal{S}^{\circ}_{2n} \quad \text{and} \quad \pi_{2n+1} = \pi_{2n-1}\mathcal{S}^{\circ}_{2n+1}. \tag{2.9}$$

For $0 \leqslant p \leqslant n$, we define the associated inhomogeneous semigroups by composition,

$$\mathcal{S}^{\circ}_{2p,2n} = \mathcal{S}^{\circ}_{2p,2(n-1)}\mathcal{S}^{\circ}_{2n} \quad \text{and} \quad \mathcal{S}^{\circ}_{2p+1,2n+1} = \mathcal{S}^{\circ}_{2p+1,2n-1}\mathcal{S}^{\circ}_{2n+1},$$

with identity operators on the diagonal. This representation highlights that Sinkhorn dynamics belong to the class of time-varying Markov chains sharing common invariant measures at each step.

Remark 2.1. *Each kernel $\mathcal{S}^{\circ}_{2n+1} := \mathcal{S}_{2n}\mathcal{S}_{2n+1}$ corresponds to a two-block Gibbs sampler with target distribution*

$$\mathcal{P}_{2n}(d(x,y)) = \lambda_U(dx)\ \mathcal{S}_{2n}(x,dy) = \pi_{2n}(dy)\ \mathcal{S}_{2n+1}(y,dx)$$

For $n = 0$, this recovers classical proximal and data-augmentation samplers; see, e.g., [34, 89, 105, 106] *and related works* [20, 21, 56, 64, 74, 76, 83].

2.3. **Some stability theorems.** In contrast with Kantorovich criteria or related Wasserstein-type semi-distances, even when $\mathbb{X} = \mathbb{Y}$ and the marginals coincide, $\lambda_U = \nu_V$, the identity-map deterministic coupling has infinite entropy. That is, for any reference measure $\mathcal{P}_0 = (\lambda_U \times \mathcal{Q})$ of the form (1.3) we have

$$\mathcal{P}(d(x,y)) := \lambda_U(dx)\ \delta_x(dy) \in \Pi_{U,V} \Longrightarrow \mathrm{Ent}(\mathcal{P} \mid \mathcal{P}_0) = +\infty.$$

This fact, together with (1.5), shows that the Schrödinger bridge $\mathbb{P}$ and the convergence analysis of Sinkhorn bridges $\mathcal{P}_{2n}$ crucially depend on the form of the reference transition $\mathcal{Q}$ and the transition potential function W discussed in (1.2).

Markov transitions of the form (1.2) with a bounded potential W typically arise on finite or compact spaces. For instance, Markov processes on compact manifolds, such as diffusions on bounded manifolds reflected at the boundary, have Markov transitions of the form (1.2) with a uniformly bounded transition potential W that depends on the time parameter [6, 7, 84, 103]. The convergence analysis of Sinkhorn bridges in this context is often developed using Hilbert Hilbert projective metrics [18, 46, 60, 79].

In the context of bounded transition potentials, the contraction principles of Markov operators presented in the article [43] yield an alternative stability analysis that also applies to any generalized Φ-entropies introduced in (1.13).

Theorem 2.2. *Assume that the transition potential W is uniformly bounded. In this case, for any Φ-entropy criteria D_Φ there is some parameters $\rho \in]0,1[$ and $c > 0$ such that for any $n \geqslant 0$ we have the contraction inequalities*

$$D_\Phi(\pi_{2n}, \nu_V) \leqslant \rho^n\ D_\Phi(\pi_0, \nu_V) \quad \text{and} \quad \|\log(d\pi_{2n}/d\nu_V)\| \leqslant c\ \rho^n. \tag{2.10}$$

The above inequalities remain valid when we replace (π_{2n}, ν_V) by (ν_V, π_{2n}). Analogous bounds hold for (λ_U, π_{2n+1}). Moreover,

$$\mathrm{Ent}(\mathbb{P} \mid \mathcal{P}_{2n}) \leqslant c\ \rho^n. \tag{2.11}$$

The estimates (2.10) are a consequence of the sandwich-type contraction inequalities stated in Theorem 4.1. A more refined version of the entropy estimate (2.11) in terms of Dobrushin contraction coefficients is provided in Corollary 4.2

We underline that even on totally bounded state spaces such as $\mathbb{X} = \mathbb{Y} =]0,1[$ the transition potential function W may be unbounded. Indeed, consider the reference Markov transition defined for any $x \in]0,1[$ by

$$\mathcal{Q}(x,dy) = \frac{1}{q(x)}\ (1 - |x-y|)\ 1_{]0,1[}(y)\ dy \quad \text{with} \quad \frac{1}{4} \leqslant q(x) = \frac{1}{2} - x(1-x) \leqslant \frac{1}{2}. \tag{2.12}$$

In this context, we have

$$W(x,y) = -\log((x \wedge y) + (1 - (x \vee y))) + \log q(x).$$

The function $W(x,y)$ is locally bounded on $]0,1[^2$ but tends to ∞ on boundaries. For instance, we obtain

$$W\left(\frac{1}{n}, 1 - \frac{1}{n}\right) \geqslant \log\frac{n}{2} - 2\log 2 \stackrel{n\to\infty}{\longrightarrow} +\infty.$$

This rather simple model does not satisfy the boundedness condition of Theorem 2.2. To handle locally bounded transition potentials, consider the functions

$$\mathcal{U}_\delta := \delta U - W^V \quad \text{and} \quad \mathcal{V}_\delta := \delta V - W_U \tag{2.13}$$

with the integrated costs

$$W_U(y) := \int \lambda_U(dx)\ W(x,y) \quad \text{and} \quad W^V(x) := \int \nu_V(dy)\ W(x,y). \tag{2.14}$$